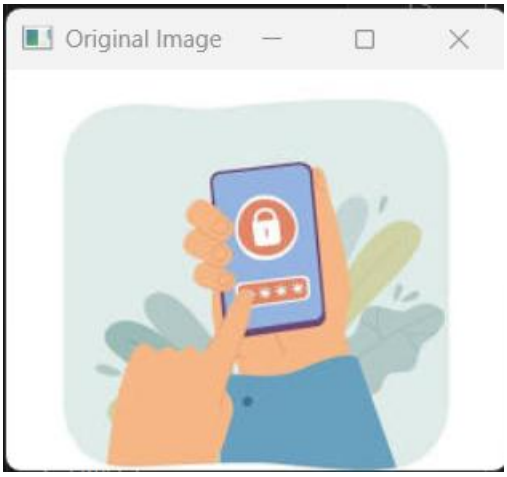
```
# Task1
import cv2

# Part 1 – Image Processing Module
image = cv2.imread("student_id.jpg")

if image is None:
    print("Error: student_id.jpg not found!")
else:
    print("Image Loaded Successfully!")

    cv2.imshow("Original Image", image)

    cv2.imwrite("backup.jpg", image)

    resized = cv2.resize(image, (300, 300))
    print("Image Resized to 300x300")
    cv2.imshow("Resized Image", resized)

    cropped = resized[50:200, 100:250]
    print("Cropped Region Displayed")
    cv2.imshow("Cropped Image", cropped)

    gray = cv2.cvtColor(resized, cv2.COLOR_BGR2GRAY)
    print("Converted to Grayscale")
    cv2.imshow("Grayscale Image", gray)

    hsv = cv2.cvtColor(resized, cv2.COLOR_BGR2HSV)
    print("Converted to HSV")
    cv2.imshow("HSV Image", hsv)

    blur = cv2.GaussianBlur(resized, (15, 15), 0)
    print("Blur Applied")
    cv2.imshow("Blurred Image", blur)
```

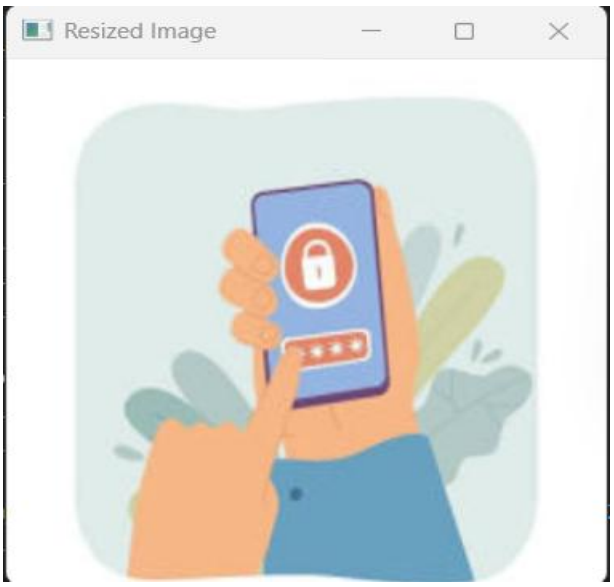
```
    print("Image Processing Module Completed!")

    cv2.waitKey(0)
    cv2.destroyAllWindows()

# Part 2 – Video Processing Module
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Cannot open camera")
else:
    print("Camera Started...")
    print("Press 'q' to Exit")

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        cv2.imshow("Smart Campus Camera", frame)

        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

    cap.release()
    cv2.destroyAllWindows()
    print("Camera Closed Successfully")
```

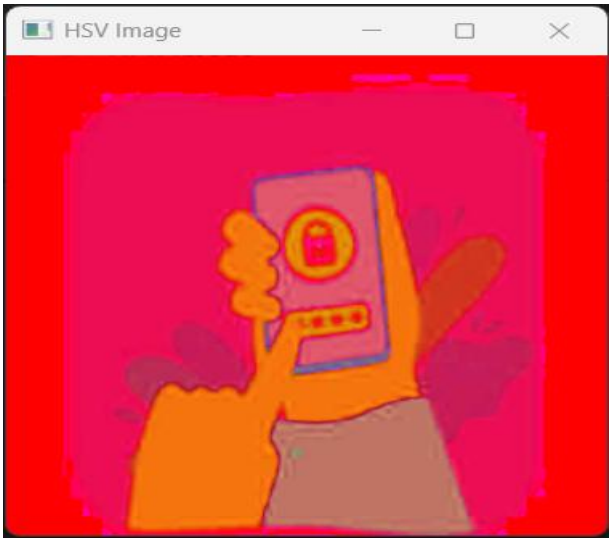
```
# Task2
import pyautogui
import time

time.sleep(2)

# PART 1 – Application Launch Automation
print("Launching Calculator...")
pyautogui.hotkey('win', 'r')
time.sleep(1)
pyautogui.write('calc')
pyautogui.press('enter')
time.sleep(2)
print("Calculator Opened Successfully")

# PART 2 – Automated Calculation
print("Performing Automated Calculation...")
pyautogui.write('1234+5678')
pyautogui.press('enter')
time.sleep(1)

screenshot = pyautogui.screenshot(region=(500, 300, 400, 300))
screenshot.save("calculator_result.png")

print("Result Captured Successfully")

# PART 3 – Mouse Control Simulation
print("Mouse Movement Started")

pyautogui.moveTo(300, 300, duration=1)
pyautogui.click()

pyautogui.doubleClick()

pyautogui.rightClick()
```

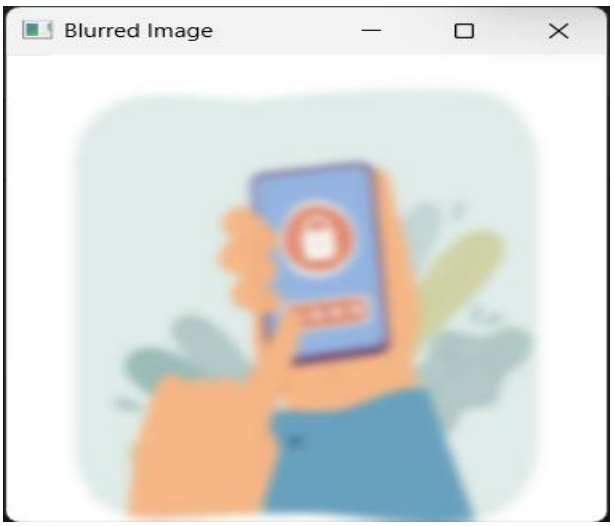
```
pyautogui.dragTo(600, 400, duration=1)

print("Click Actions Completed")
print("Drag Operation Completed")

# PART 4 – Typing & Keyboard Automation
print("Opening Notepad...")
pyautogui.hotkey('win', 'r')
time.sleep(1)
pyautogui.write('notepad')
pyautogui.press('enter')
time.sleep(2)

print("Typing Report...")
pyautogui.write("Daily Automation Report Generated")
pyautogui.press('enter')
pyautogui.write("Date: 03-03-2026")

pyautogui.hotkey('ctrl', 's')
time.sleep(1)
pyautogui.write('report.txt')
pyautogui.press('enter')

print("Report Saved Successfully")

# PART 5 – Screenshot Automation
full = pyautogui.screenshot()
full.save("full_screen.png")
print("Full Screen Captured")

partial = pyautogui.screenshot(region=(100, 100, 500, 400))
partial.save("partial_screen.png")
print("Partial Screen Captured")

print("Screenshots Saved Successfully")
```

Outputs:

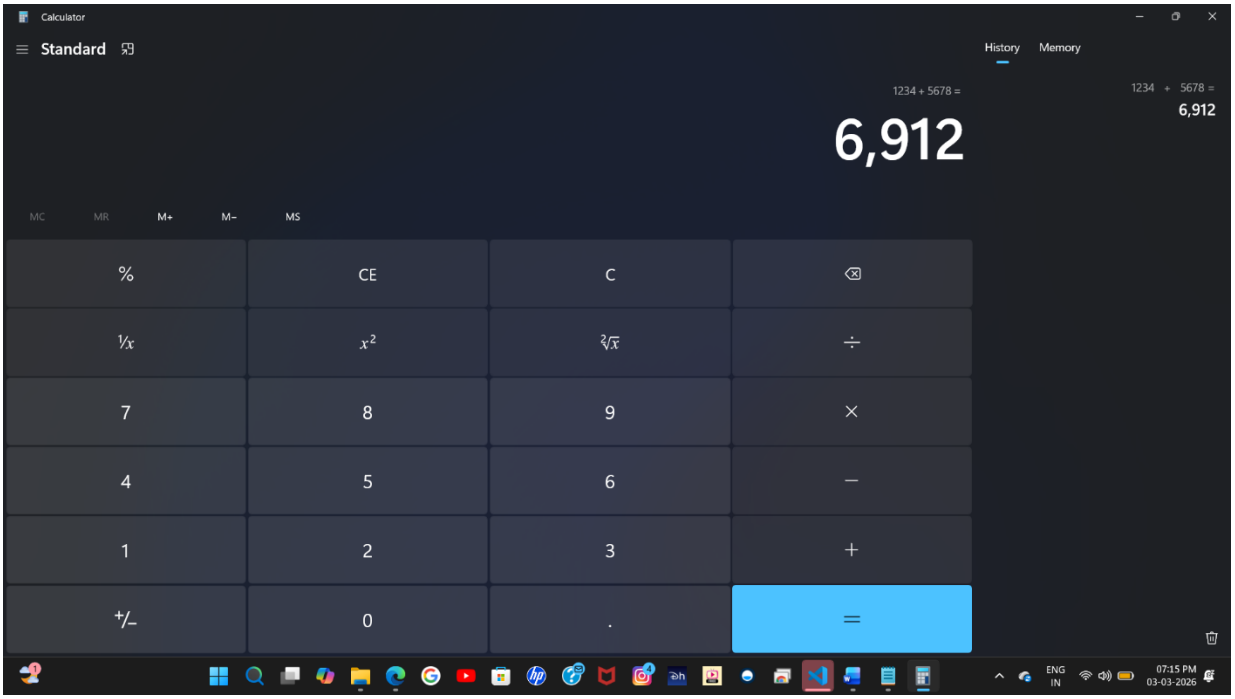


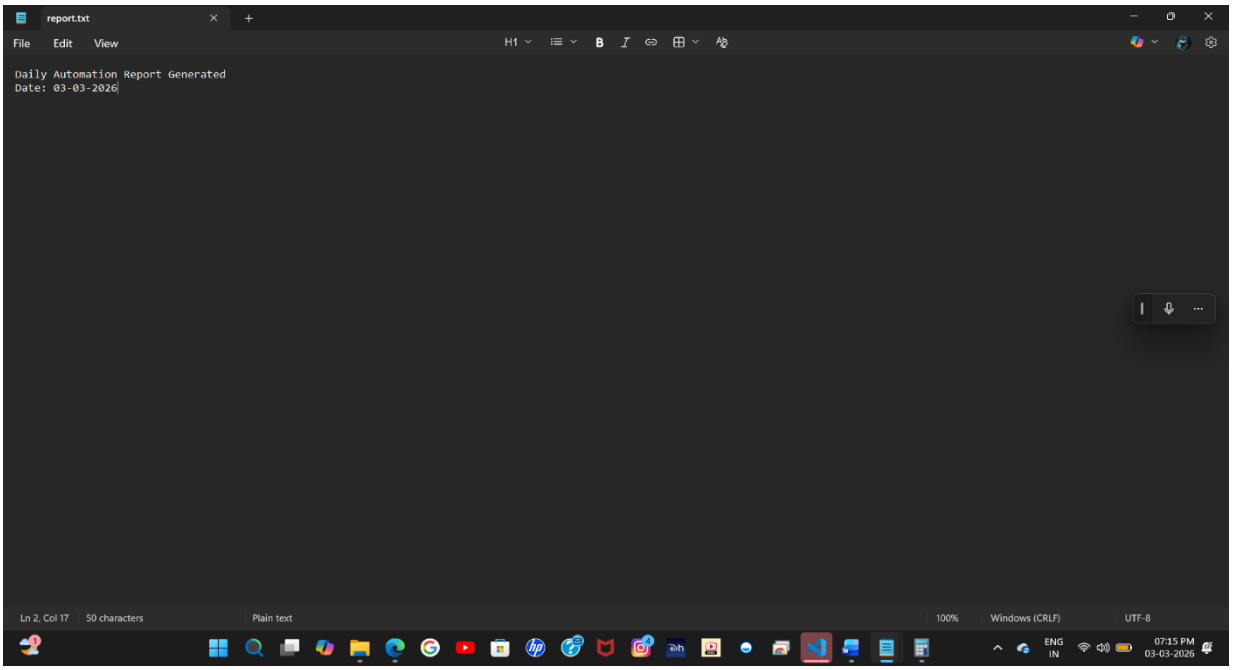







```
(.venv) PS D:\Internship\Day26> python task1.py
Image Loaded Successfully!
Image Resized to 300x300
Cropped Region Displayed
Converted to Grayscale
Converted to HSV
Blur Applied
Image Processing Module Completed!
Camera Started...
Press 'q' to Exit
Camera Closed Successfully
```






```
(.venv) PS D:\Internship\Day26> python task2.py
Launching Calculator...
Calculator Opened Successfully
Performing Automated Calculation...
Result Captured Successfully
Mouse Movement Started
Click Actions Completed
Drag Operation Completed
Opening Notepad...
Typing Report...
Report Saved Successfully
Full Screen Captured
Partial Screen Captured
Screenshots Saved Successfully
```